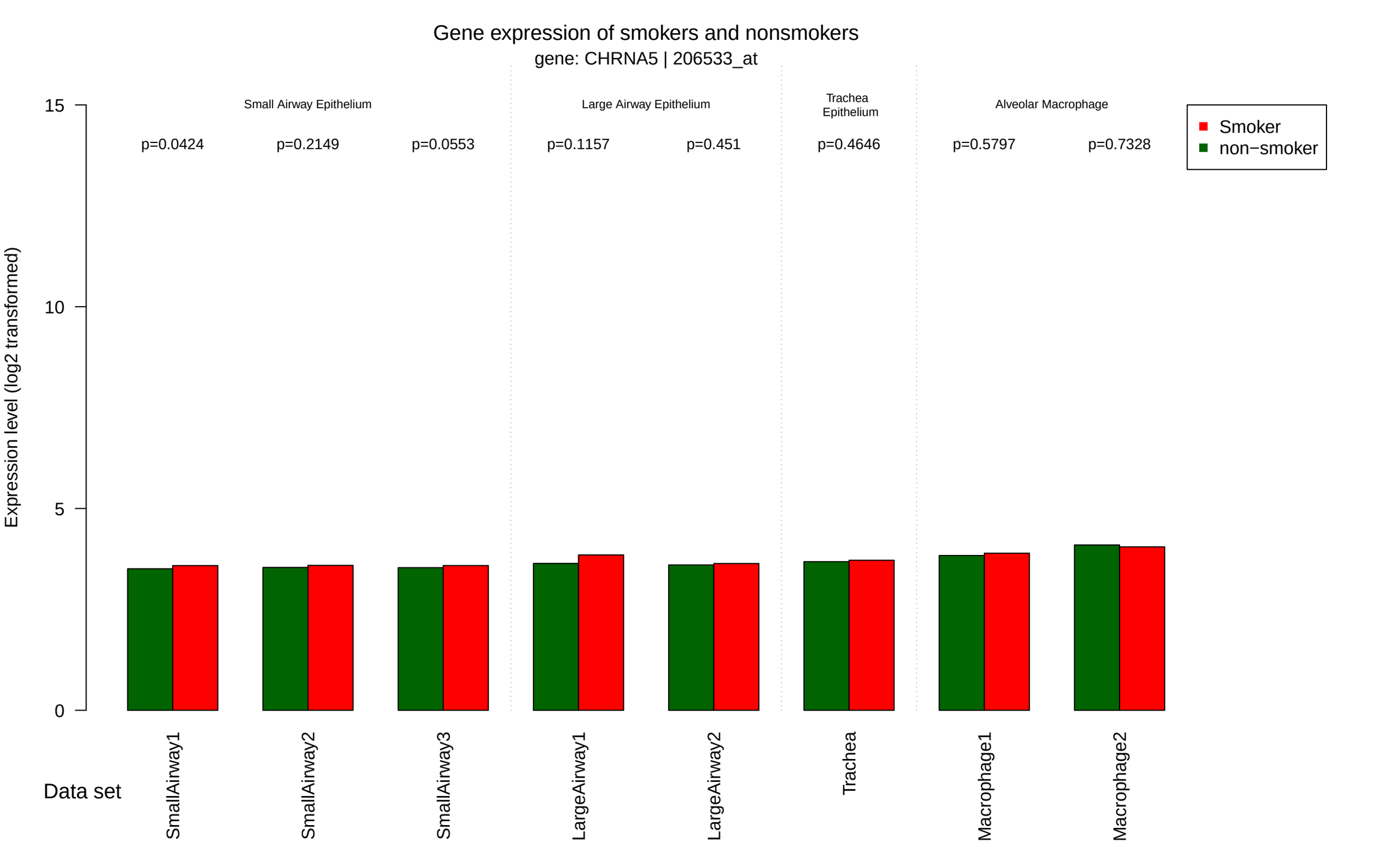
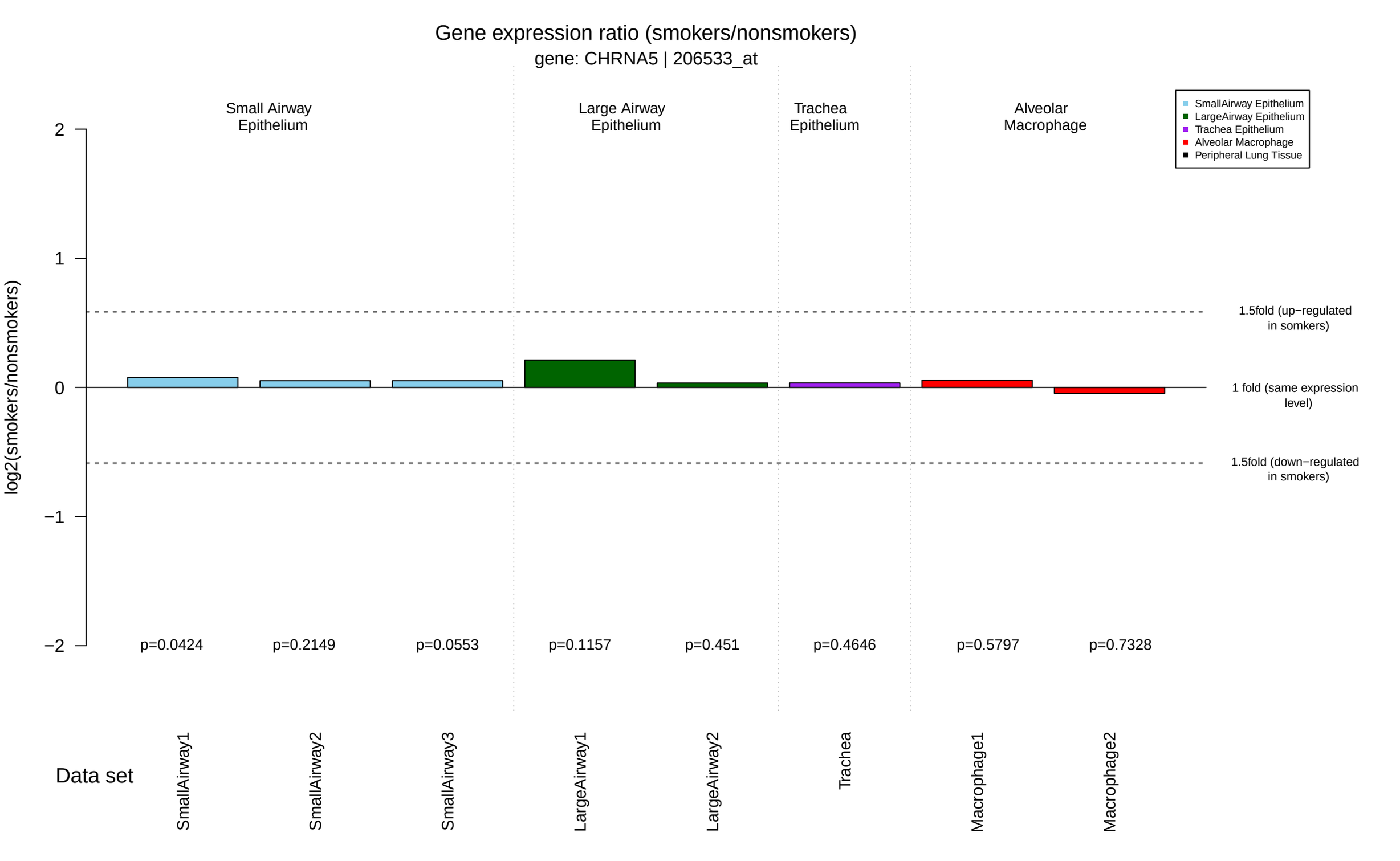
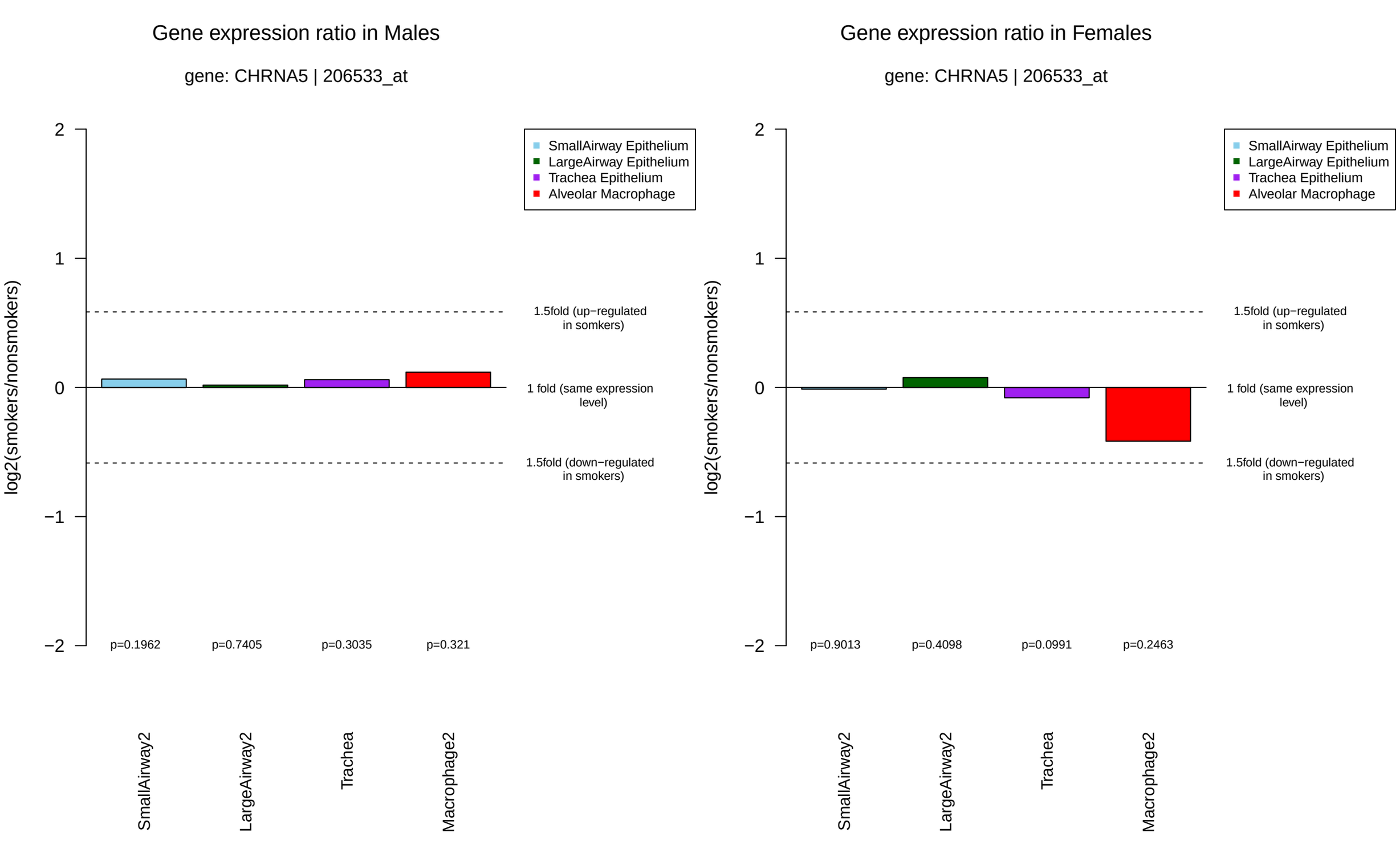
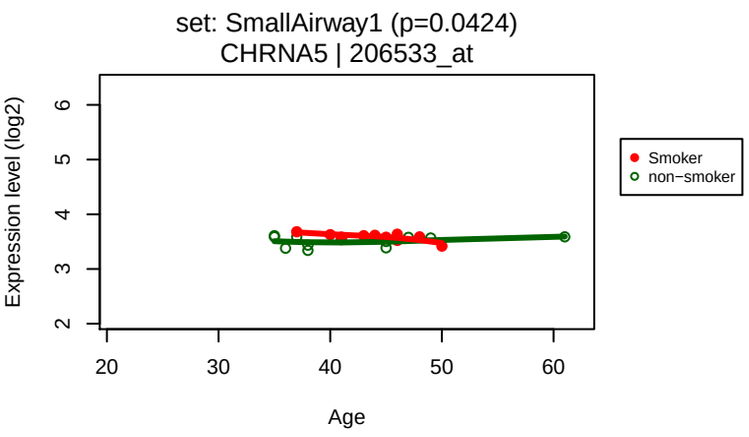
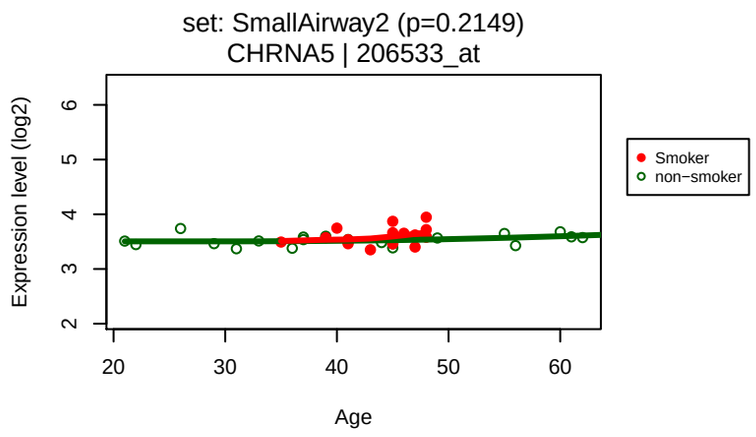
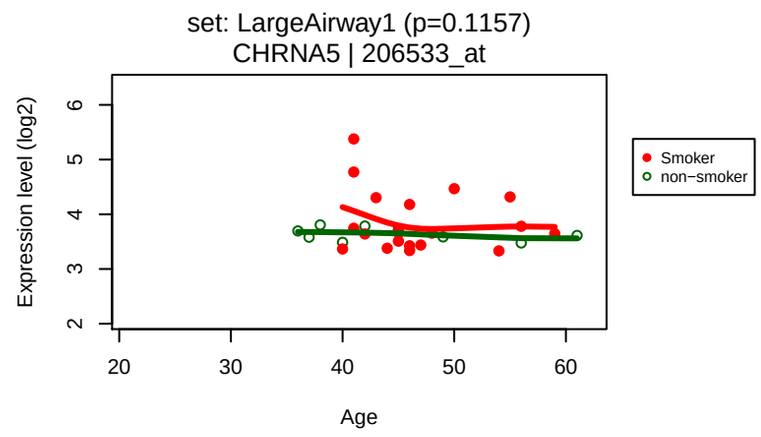

Gene expression of smokers and nonsmokers
gene: CHRNA5 | 206533_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0424
p=0.2149
p=0.0553
p=0.1157
p=0.451
p=0.4646
p=0.5797
p=0.7328
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

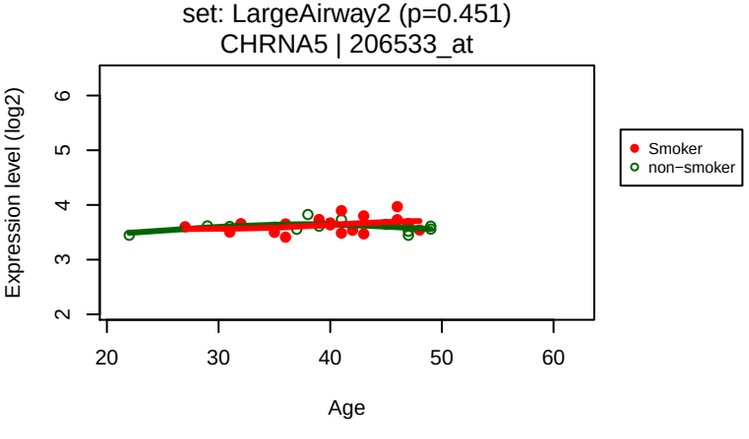

Gene expression ratio (smokers/nonsmokers)
gene: CHRNA5 | 206533_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0424
p=0.2149
p=0.0553
p=0.1157
p=0.451
p=0.4646
p=0.5797
p=0.7328
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

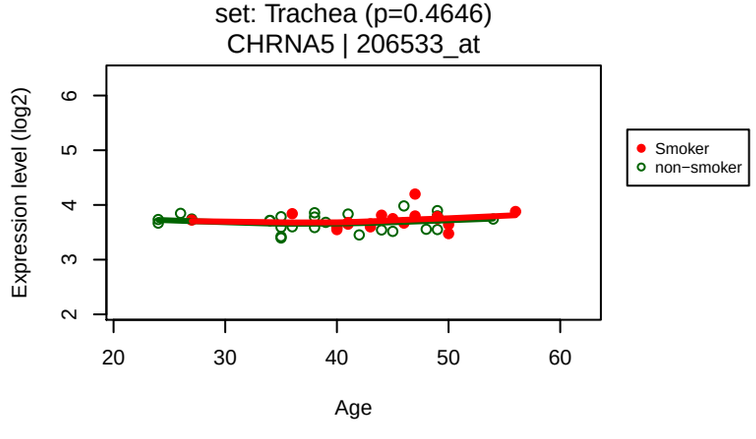

Gene expression ratio in Males
gene: CHRNA5 | 206533_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1962
p=0.7405
p=0.3035
p=0.321
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CHRNA5 | 206533_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9013
p=0.4098
p=0.0991
p=0.2463
SmallAirway2
LargeAirway2
Trachea
Macrophage2

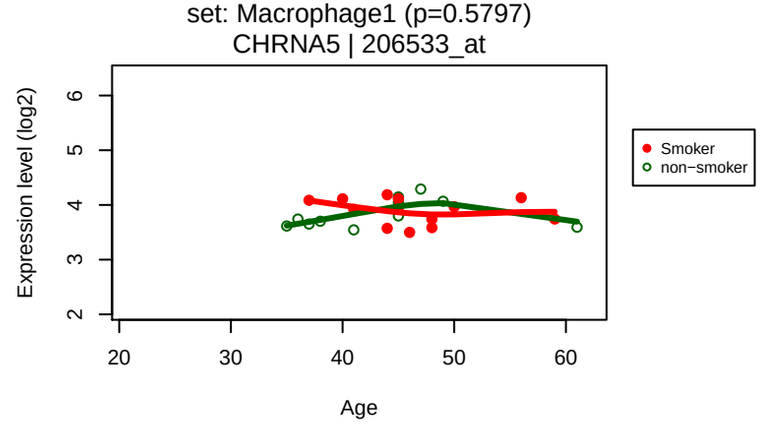

set: SmallAirway1 (p=0.0424)
CHRNA5 | 206533_at
Expression level (log2)
Age
Smoker
non−smoker

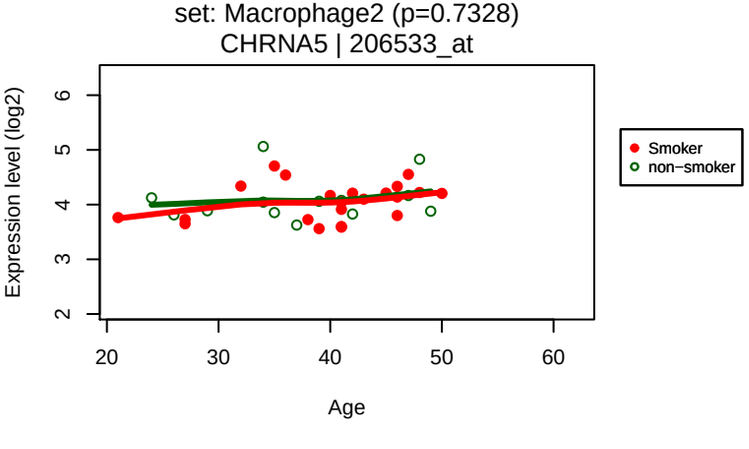

set: SmallAirway2 (p=0.2149)
CHRNA5 | 206533_at
Expression level (log2)
Age
Smoker
non−smoker

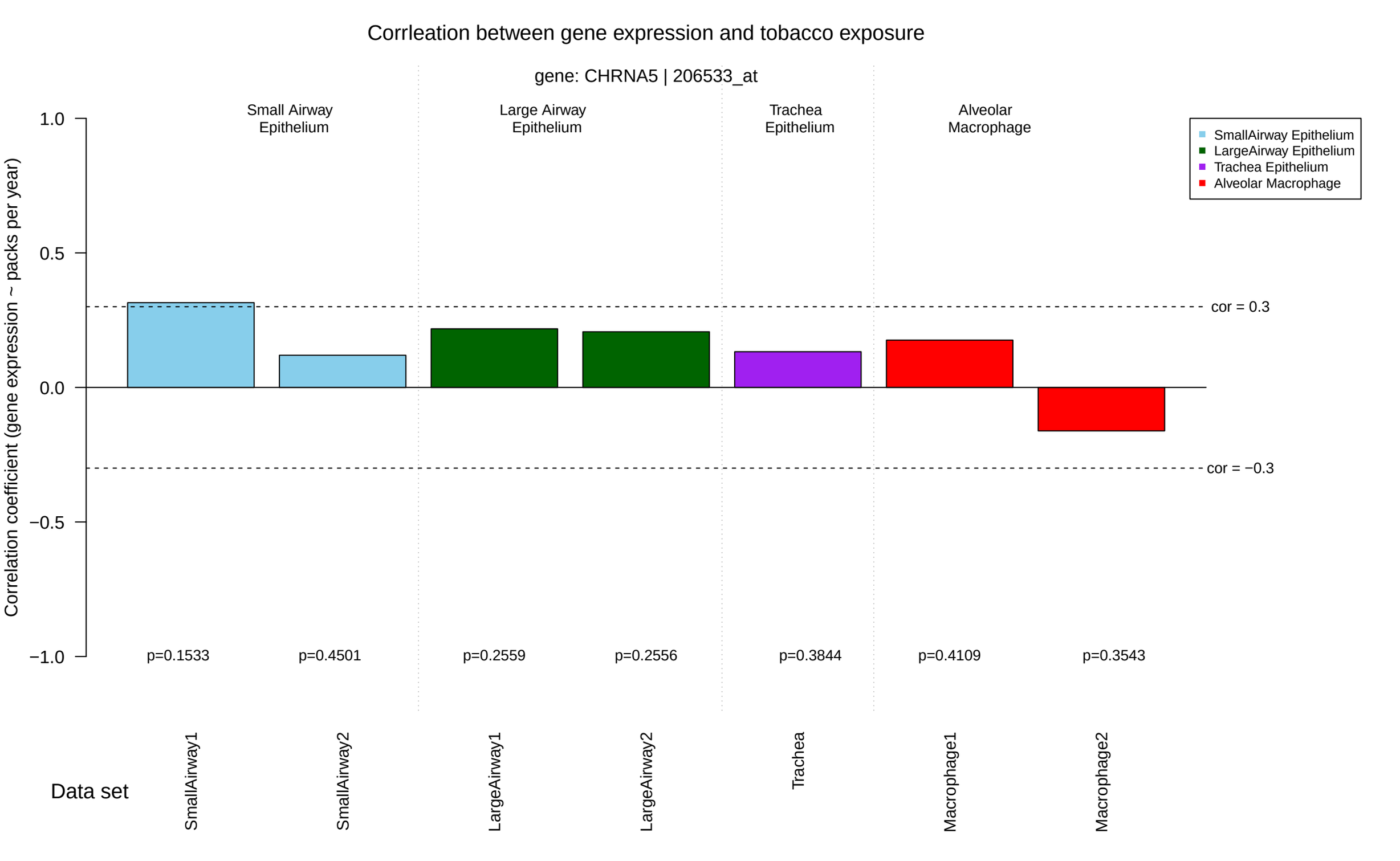

set: LargeAirway1 (p=0.1157)
CHRNA5 | 206533_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.451)
CHRNA5 | 206533_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4646)
CHRNA5 | 206533_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5797)
CHRNA5 | 206533_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7328)
CHRNA5 | 206533_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CHRNA5 | 206533_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1533
p=0.4501
p=0.2559
p=0.2556
p=0.3844
p=0.4109
p=0.3543
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2